SAI VENKATA HARSHINI RAYAVARAPU

Cleveland, OH | 614-674-3964 | rayavarapu.harshini@gmail.com | linkedin.com/in/sai-venkata-harshini-rayavarapu | github.com/Hercodes-ux | hercodes-ux.github.io | medium.com/@rayavarapu.harshini

SUMMARY

MS Computer Science graduate (Cleveland State, May 2026) with 14 months at Accenture building banking data pipelines in SQL and end-to-end Python systems built during the degree. Hands-on data engineering, applied analytics, machine learning, and AI-enabled workflows — a retrieval-and-ranking pipeline with an exact flat inner-product search index, a multi-agent SQL system with automated query validation, and a causal-inference lift framework. Strengths are verified, reproducible analysis and systems that fail safe instead of returning a confident wrong answer.

TECHNICAL SKILLS

Languages: Python (PyTorch, Pandas, NumPy, Scikit-Learn), SQL

ML & Data: Two-Tower Retrieval, Embeddings, Exact Flat Inner-Product Search, Causal Inference, Propensity Score Matching, Multi-Agent Orchestration (LangGraph)

Data & Tools: FastAPI, DuckDB, Plotly, Power BI (DAX, Power Query), Git, GitHub Actions

Practices: Automated Testing (pytest), CI/CD, Debugging Against Ground Truth, Query Validation, Data Quality

PROJECTS

NeuralStream — Neural Retrieval & Ranking System

PyTorch | FAISS (Exact Flat Inner-Product) | FastAPI

- Built a two-stage retrieval-and-ranking pipeline: a two-tower network embeds users and items in a shared 64-D space.
- Built an exact FAISS flat inner-product vector-search index over a 1M-item catalog (embeddings precomputed offline), serving top-K retrieval at sub-50ms p95 latency.
- Verified index results score-for-score against PyTorch ground truth when debugging, and evaluated head and tail items separately as a guard against popularity bias in positives-only training data.

Starbit Engine — Autonomous Multi-Agent Data Intelligence

LangGraph | Llama 3.3 (Groq) | DuckDB

- Architected a LangGraph multi-agent system that plans, writes, validates, and visualizes SQL against DuckDB without human intervention.
- Added a critic agent that catches hallucinated or invalid SQL and triggers bounded recursive retries, reducing failed queries on multi-table schemas.
- Designed the system for verifiability: no query executes without passing validation, trading latency for reliability.

Causal Analytics — Incremental Lift & ROI Framework

Scikit-Learn | Statsmodels | Pandas

- Built a causal inference pipeline measuring true incremental lift of product features, constructing counterfactual control groups via Propensity Score Matching (KNN) to isolate treatment effect from self-selection bias.
- Validated the estimator on a simulated campaign dataset with known ground truth before applying it to observational data.
- Communicated lift findings as documented, reproducible analysis rather than correlational estimates.

PROFESSIONAL EXPERIENCE

Accenture Pvt Ltd — Bengaluru, India

Application Development Associate | October 2022 - December 2023

- Built and maintained high-volume banking data pipelines in Mainframe SQL: ETL jobs and data-quality checks keeping downstream reporting consistent.
- Automated reconciliation steps in SQL, replacing recurring manual effort in operational reporting and improving reporting reliability.
- Tuned SQL execution plans and indexing to reduce data retrieval latency across reporting queries.
- Re-engineered legacy reporting logic into modern, auditable SQL with documented data-quality controls.

EDUCATION

Cleveland State University — Cleveland, OH

Master of Science in Computer Science | GPA: 3.4 / 4.0 | January 2024 - May 2026

Coursework: Artificial Intelligence, Machine Learning, Data Science, Network Security & Privacy

Vignan's Institute of Information Technology — Visakhapatnam, India

Bachelor of Technology in Computer Science & Engineering | GPA: 8.3 / 10 | May 2022

CERTIFICATIONS

OpenAI Academy — Agents and Workflows (2026) | Databricks Accreditation — AI Agent Fundamentals & Databricks Fundamentals (2026) | AWS — Fundamentals of ML & AI (2026)

Work Authorization: F-1 OPT work-authorized; eligible for 24-month STEM OPT extension (36 months total); no initial sponsorship required